Practice Questions

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Score: 67% Passing Score: 80%

Question 1: ✕ Incorrect

What is the speed of an OC-3 connection?

- ~~622 Mbps~~
- 10 Mbps
- 155 Mbps
- 100 Mbps

EXPLANATION

Optical Carrier (OC) is used to specify the speed of fiber optic networks conforming to the SONET standard. Common OC speeds are:

- OC-1 = 51.85 Mbps
- OC-3 = 155.52 Mbps
- OC-12 = 622.08 Mbps
- OC-24 = 1.244 Gbps
- OC-48 = 2.488 Gbps
- OC-192=9.952 Gbps

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 2: ✔ Correct

To access the internet through the PSTN, what kind of connectivity device must you use?

- Switch
- TDM
- DTE
- CSU/DSU
- ➡ Modem

EXPLANATION

To establish a connection to the internet through the public telephone network (PSTN/POTS) you must use a modem (modulator/demodulator), which converts digital PC data into analog signals that can be transmitted through standard telephone lines.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 3: ✔ Correct

Which of the following technologies uses variable-length packets, adds labels to packets as they enter the WAN cloud, and uses the labels to switch packets and prioritize traffic?

- ➡ MPLS
- Frame relay
- SONET
- ATM
- ISDN

EXPLANATION

MPLS is a WAN data classification and data carrying mechanism.

- MPLS is a packet switching technology that supports variable-length frames.
- MPLS adds a label to packets between the existing Network and Data Link layer formats. Labels are added when the packet enters the MPLS network and removed when the packet exits the network.
- Information in the label is used to switch the packet through the MPLS network to the destination.
- MPLS labels can identify the route or even the network type to use. MPLS labels are often used to provide different classes of service for data streams.

ATM is a WAN communication technology that uses labels but has fixed-length cells of 53 bytes. Frame relay is a protocol used to connect to a WAN over dedicated (leased) lines. ISDN is a WAN technology that provides increased bandwidth within the local loop. SONET is a standard for networking over an optical medium. SONET is classified as a transport protocol, in that it can carry other types of traffic such as ATM, Ethernet, and IP. Most PSTN networks use SONET within the long-distance portion of the PSTN network.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 4: ✔ Correct

Which of the following are characteristics of MPLS? (Select two.)

- [ ] Uses a committed information rate (CIR)
- [x] Supports variable-length data units
- [ ] Uses fixed-length data units
- [x] Adds labels to data units
- [ ] Is a local loop technology

EXPLANATION

MPLS is a WAN data classification and data carrying mechanism.

- MPLS is a packet switching technology that supports variable-length data units.
- MPLS adds a label to packets between the existing Network and Data Link layer formats. Labels are added when the packet enters the MPLS network and removed when the packet exits the network.
- Information in the label is used to switch the packet through the MPLS network to the destination.
- MPLS labels can identify the route or even the network type to use. MPLS labels are often used to provide different classes of service for data streams.

ATM uses labels, but uses fixed-length cells. ISDN is a local loop technology. Frame relay uses the committed information rate (CIR) to define levels of service. MPLS service levels are provided by information within the labels.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 5: ✕ Incorrect

Which of the following are characteristics of ATM? (Select two.)

- [x] Uses fixed-length cells of 53 bytes.
- [ ] Connects to the WAN cloud with a CSU/DSU.
- [x] ~~Uses POTS in the local loop.~~
- [ ] Supports variable-length packets.
- [ ] Adds labels to data units.

EXPLANATION

ATM is a WAN communication technology originally designed for carrying time-sensitive data, such as voice and video. However, it can also be used for regular data transport.

- ATM is a packet switching technology that uses fixed-length data units called cells. Each cell is 53 bytes.
- The cell header includes labels that identify the virtual path information. ATM switches in the WAN cloud use the virtual path to switch cells within the WAN to the destination.

MPLS uses labels but supports variable-length packets. ISDN is a WAN technology that uses existing telephone lines (POTS) to connect to the WAN cloud. Both frame relay and ISDN use a CSU/DSU to connect the line to the WAN link.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 6: ✕ Incorrect

Which of the following are the the WAN service provider's responsibility to maintain? (Select three.)

- [ ] CPE
- [x] Local loop
- [x] PSE
- [x] ~~DTE~~
- [ ] CO

EXPLANATION

WAN service providers are responsible for:

- Packet switching exchange (PSE)—equipment inside the WAN cloud.
- Central office (CO)—equipment that allows access to the PSE.
- Local loop—wiring that connects the customer to the CO.

Customers are responsible for:

- Customer premises equipment (CPE)—any equipment at the customer site.
- Data terminal equipment (DTE)—devices that send local data to the WAN.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 7: ✔ Correct

Which network type divides transmitted data into smaller pieces and allows multiple communications on the network medium?

- Frame-based
- Circuit-switched
- Multiplexed
- ➡ Packet-switched
- Sliding window

EXPLANATION

A packet-switched network divides data into small units called packets. These packets are routed by their destination addresses. In a packet-switched network, multiple hosts can use the network medium at the same time. An Ethernet computer network is an example of a packet-switched network.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 8: ✔ Correct

Which network type establishes a dedicated physical connection between two hosts in order to transmit time-sensitive data?

- Multiplexed
- ➡ Circuit-switched
- Frame-based
- Packet-switched
- Sliding window

EXPLANATION

A circuit-switched network establishes a dedicated physical circuit between two hosts that need to transfer time sensitive data, such as real-time audio and video. No other network host may use the medium until the communication is complete. An example of a circuit-switched network is the public switched telephone network.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 9: ✕ Incorrect

Which of the following WAN technologies provides packet switching over high-quality digital lines at speeds greater than 1.544 Mbps?

- ~~ISDN~~
- Frame relay
- POTS
- X.25

EXPLANATION

Frame relay networks are packet-switched networks that operate at T1 or T3 speeds. Frame relay achieves faster speeds by using high-quality digital lines, which require less extensive error control than X.25 networks.

POTS stands for plain old telephone system. In the United States, ISDN networks can only offer speed up to 1.544 Mbps.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 10: ✕ Incorrect

You are implementing internet connectivity for a new start-up company. Your client will provide online storefronts for retailers. To do this, they have calculated that their internet connection must provide a data rate of at least 20–30 Mbps.

Which type of service should you implement?

- T3
- Dial-up
- ISDN
- ~~Integrated T1~~
- T1
- Fractional T1

EXPLANATION

The T3 carrier service is similar to T1; however, it provides a throughput of 44.736 Mbps. The T1 service provides a data rate of only 1.544 Mbps. T3 is the only listed service that provides a data rate fast enough for the client in this scenario.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 11: ✔ Correct

Which networking technology creates virtual links between two remote network endpoints by prefixing packets with a header containing one or more labels?

- ➡ (●) MPLS
- ( ) Frame relay
- ( ) ISDN
- ( ) Ethernet

EXPLANATION

Multiprotocol Label Switching (MPLS) creates virtual links between two remote network endpoints by prefixing packets with a header containing one or more labels. MPLS determines routing using the contents of each packet's label. This allows the creation of an endpoint-to-endpoint virtual circuit across a variety of network media.

Ethernet, ISDN, and frame relay WAN networks do not use labels.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 12: ✔ Correct

Which of the following describe the channels and data transfer rates used for ISDN BRI? (Select two.)

- [ ] One D channel operating at 64 Kbps
- ➡ [x] Two B channels operating at 64 Kbps each
- ➡ [x] One D channel operating at 16 Kbps
- [ ] 23 B channels operating at 64 Kbps each

EXPLANATION

ISDN BRI uses two B channels operating at 64 Kbps each and one D channel operating at 16 Kbps. It is often called 2B + 1D.

ISDN PRI uses 23 B channels (at 64 Kbps) and one D channel (at 64 Kbps). It is also called 23B + 1D.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 13: ✔ Correct

You are traveling throughout North America to many metropolitan and rural areas.

Which single form of internet connectivity provides the greatest potential connectivity wherever you travel?

- Broadband cable
- ISDN
- PSTN
- DSL

EXPLANATION

Network access using a modem over the telephone company network (PSTN) is not the fastest method for internet connectivity. However, it has the advantage of being available virtually anywhere that regular voice-grade communications are available.

Broadband cable is dependent on service offerings from the regional cable television company, which does not have as great a presence as the telephone company. To use broadband cable, the service must be added to the cable TV lines. DSL and ISDN are offered through the telephone company; however, they are not available in all service areas. And even when available, they require that the subscriber be within a certain proximity of telephone company equipment.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 14: ✔ Correct

What must you install between your network and a T1 line for your network to use the T1 line?

- Bridge
- Transceiver
- Gateway
- ➡ CSU/DSU

EXPLANATION

A CSU/DSU is required for a network to send signals on a T1 line.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 15: ✔ Correct

Which of the following devices is used on a WAN to convert synchronous serial signals into digital signals?

- Modem
- ➡ CSU/DSU
- IDS
- Proxy

EXPLANATION

A CSU/DSU is a device that converts the signal received from the WAN provider into a signal that can be used by equipment at the customer site. A CSU/DSU is composed of two separate devices.

- The CSU terminates the digital signal and provides error correction and line monitoring.
- The DSU converts the digital data into synchronous serial data for connection to a router.

The CSU/DSU might be two separate devices, one combined device, or integrated into a router.

A modem converts digital signals to analog signals. A proxy server is a type of firewall that can filter based on upper-layer data. An intrusion detection system (IDS) is a special network device that can detect attacks and suspicious activity.

REFERENCES

LabSim for Network Pro, Section 11.1.

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